7.1 INTRODUCTION

Based on type, the global microbiome sequencing services market is segmented into amplicon sequencing, whole-genome sequencing, shotgun sequencing, and other types (including de novo sequencing, metagenomic sequencing, and metatranscriptomic sequencing). The amplicon sequencing segment accounted for the largest market share of 41.0% in 2022. The large share of this segment can be attributed to the high adoption of 16S rRNA microbiome sequencing and targeted sequencing. Key market players/service providers have built an innovative portfolio around these services, including sample preparation and data analysis solutions.

TABLE 29 MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	91.80	104.54	118.98	135.54	154.55	176.84	203.06	234.18	14.5%
Whole-genome Sequencing	47.29	53.48	60.45	68.39	77.44	87.98	100.31	114.85	13.7%
Shotgun Sequencing	59.09	67.70	77.52	88.83	101.89	117.27	135.44	157.10	15.2%
Other Types	21.85	24.46	27.36	30.62	34.29	38.52	43.41	49.11	12.4%
Total	**220.03**	**250.17**	**284.31**	**323.39**	**368.16**	**420.61**	**482.21**	**555.24**	**14.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

7.2 AMPLICON SEQUENCING

7.2.1 HIGH ACCURACY OF 16S AMPLICON SEQUENCING TO DRIVE ADOPTION

This segment is inclusive of 16S rRNA sequencing, ITS sequencing, and targeted sequencing services. Amplicon sequencing is a highly targeted approach that allows researchers to analyze genetic variation in specific regions of a genome. The ultra-deep sequencing of PCR products (amplicons) allows efficient variant identification and characterization. 16S rRNA sequencing is the most common technique utilized for metagenomics studies. In this technique, the 16S rRNA genetic markers present in a bacterial genome are sequenced to study bacterial phylogeny and taxonomy from complex microbiomes or environments that are difficult or impossible to analyze. The presence of 16S rRNA genetic markers as a multigene family or operons in almost all bacteria and the large size of the 16S rRNA gene (~1,500 bp) makes it the most preferred metagenomic sequencing technique among end users.

The demand for targeted sequencing is anticipated to increase in metagenomic research activities. The targeted amplicon sequencing method involves sequencing a phylogenetically informative marker to identify organisms in community samples. The marker must be present across all samples with expected organisms to enhance taxonomic profiling through the amplification of the primer. A number of different markers are commonly used and vary by the taxonomy of interest, but the most commonly used targeted amplicon approaches include the 16S rRNA gene for bacterial samples, the 18S for eukaryotes, and ITS for fungal samples.

TABLE 30 AMPLICON SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	52.63	59.60	67.44	76.40	86.62	98.54	112.50	128.99	13.8%
Europe	28.39	32.47	37.11	42.44	48.59	55.82	64.35	74.51	15.0%
Asia Pacific	7.79	9.16	10.75	12.63	14.83	17.47	20.63	24.46	17.9%
Latin America	1.77	1.97	2.20	2.45	2.72	3.04	3.40	3.82	11.7%
Middle East and Africa	1.22	1.35	1.48	1.63	1.79	1.97	2.17	2.40	10.1%
Total	**91.80**	**104.54**	**118.98**	**135.54**	**154.55**	**176.84**	**203.06**	**234.18**	**14.5%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 31 NORTH AMERICA: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	47.70	54.05	61.22	69.40	78.74	89.64	102.42	117.52	13.9%
Canada	4.93	5.54	6.22	7.00	7.88	8.90	10.08	11.48	13.0%
Total	**52.63**	**59.60**	**67.44**	**76.40**	**86.62**	**98.54**	**112.50**	**128.99**	**13.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 32 EUROPE: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	6.16	7.06	8.08	9.26	10.62	12.22	14.11	16.37	15.2%
UK	5.50	6.34	7.29	8.40	9.68	11.20	13.00	15.15	15.7%
France	4.34	4.97	5.68	6.50	7.45	8.56	9.87	11.43	15.0%
Italy	1.22	1.40	1.59	1.82	2.08	2.38	2.74	3.16	14.7%
Spain	0.94	1.07	1.22	1.39	1.58	1.81	2.08	2.40	14.5%
Rest of Europe	10.22	11.64	13.24	15.08	17.18	19.65	22.55	25.99	14.4%
Total	**28.39**	**32.47**	**37.11**	**42.44**	**48.59**	**55.82**	**64.35**	**74.51**	**15.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 33 ASIA PACIFIC: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	3.30	3.88	4.55	5.34	6.27	7.39	8.72	10.34	17.8%
Japan	1.53	1.81	2.14	2.53	2.98	3.53	4.20	5.00	18.5%
India	0.19	0.23	0.27	0.31	0.37	0.43	0.51	0.60	17.7%
Rest of Asia Pacific	2.76	3.24	3.79	4.44	5.20	6.11	7.20	8.51	17.6%
Total	**7.79**	**9.16**	**10.75**	**12.63**	**14.83**	**17.47**	**20.63**	**24.46**	**17.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 34 LATIN AMERICA: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.83	0.93	1.03	1.15	1.28	1.43	1.61	1.80	11.8%
Rest of Latin America	0.94	1.04	1.16	1.29	1.44	1.61	1.80	2.02	11.7%
Total	**1.77**	**1.97**	**2.20**	**2.45**	**2.72**	**3.04**	**3.40**	**3.82**	**11.7%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

7.3 WHOLE-GENOME SEQUENCING

7.3.1 ACCURATE REFERENCE GENOMES GENERATED FOR MICROBIAL IDENTIFICATION TO BOOST MARKET

Microbial whole-genome sequencing (WGS) has proved to be an important tool for mapping the genomes of novel organisms, finishing genomes of known organisms, or comparing genomes across multiple samples. This technology further helps researchers in generating accurate reference genomes for microbial identification and other comparative genomic studies. For functional studies of the microbiome, WGS metagenomic approaches are more commonly utilized. This is because WGS is affordable and particularly because of an array of computational tools to analyze microbiome sequencing data.

The advantages offered by this technique have resulted in increased adoption among scientists/researchers. As compared to other traditional genomic analysis methods, NGS-based microbial whole-genome sequencing eliminates labor-intensive cloning steps, saves time, and simplifies the workflow. This technique also helps to identify low-frequency variants and genome rearrangements that may be missed or are too expensive to identify using other methods.